

HEALTHCARE OPERATIONS ANALYTICS

CT Imaging Turnaround Time & Radiology Workflow Optimization

Twenty thousand CT examinations, taken apart stage by stage, to find out where the time actually goes. Everyone assumed it was the scanner. It was not.

A note on the data before you read further

This project uses a synthetic dataset. There are no real patients here, no real clinicians, and no real hospital records. Nothing in it was derived from, sampled from, or de-identified out of an actual health record system.

The data was generated on purpose to behave like a real radiology department, including the messy parts. Every finding below describes this synthetic dataset and should not be read as a claim about any real hospital.

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Contents

Executive Summary	3
1. The Problem	5
2. Data and Method	6
3. What Was Wrong With the Data	8
3.1 The load that lied	8
3.2 Everything else that was wrong with it	8
3.3 Two decisions worth defending	9
3.4 Where all 20,075 rows ended up	9
4. What the Data Said	10
4.1 Two stages cause 86% of the variation	10
4.2 Priority works, exactly where you would hope	11
4.3 A finding I had to throw away	11
4.4 The difference that was not worth reporting	12
4.5 The SLA was measuring the wrong thing	12
4.6 The same answer, arrived at a different way	12
5. The Dashboard	14
5.1 The data model behind it	16
6. What I Would Tell the Department	17
7. What This Analysis Cannot Tell You	18
8. Reproducing the Work	18

Executive Summary

A radiology department knows something is wrong. CT results are unpredictable. One patient gets a scan and a signed report inside two hours; another waits half a day for the same exam. Nobody can say why, because nobody has ever measured the workflow in pieces.

Everyone assumes it is the scanner. It usually is.

This analysis took 20,000 CT examinations apart, stage by stage, and let the data decide. One condition shaped the whole project: no bottleneck was assumed in advance. That constraint is the reason it reached a conclusion nobody expected.

What the data said

Two of the six workflow stages cause 86% of all variation in turnaround time. Waiting for an order to be scheduled accounts for 50% of it. Waiting for a report to be written accounts for another 35%.

The three stages inside the imaging suite, meaning patient wait, the scan itself, and the handoff to reporting, take 24% of the average turnaround and produce 4% of the variance. They are quick, and more importantly they are consistent. Whatever else is going wrong, the imaging suite runs like a metronome.

The scanner is not the problem.

Turnaround is decided before the patient walks in, in how long an order sits waiting to be scheduled, and after they walk out, in how long a report sits waiting to be signed. The machine in the middle is doing its job.

What the analysis threw away

Two apparent findings did not survive scrutiny. Both are reported as non-findings rather than quietly dropped, because saying no to a result is part of the work.

- Contrast exams look 8.8 minutes slower. Splitting by exam type shows the comparison is measuring exam mix wearing a contrast label. Within CT Angiography, the exam type that uses the most contrast and takes the longest, the difference disappears entirely.
- Scanner CT-03 is 8.1 minutes slower than CT-02. Against a standard deviation of 55 minutes, that is noise. Telling a department to investigate CT-03 on that evidence sends people looking for a problem that is not there.

The numbers

Metric	Value
Completed exams analyzed	18,981
Average turnaround	177.4 min
Median turnaround	169.0 min
90th percentile turnaround	248.0 min
Biggest source of variation	Order to Schedule, 50.4%
Second biggest	Reporting, 35.4%
Demand lost to cancellations and no-shows	3.97% of orders
SLA compliance, flat 120-minute target	12.2%

Metric	Value
SLA compliance, priority-tiered targets	51.2%

1. The Problem

CT turnaround times were inconsistent, and the department had no visibility into why. Without a breakdown of the workflow into measurable pieces, leadership could not tell whether the delay lived in scheduling, in the scan itself, or in the radiologist reading queue.

That gap has real costs. Improvement effort gets aimed at the wrong target. Staffing and equipment decisions get made on instinct. And nobody can give a referring physician an honest answer about when a result will land.

So the question was simple to state and harder to answer.

Where in the CT workflow is time actually being lost, and what explains the difference between a fast exam and a slow one?

What the project set out to do

1. Measure the time spent at every stage of the CT process, from the moment an order is placed to the moment a report is signed.
2. Work out which stage contributes most to turnaround, without deciding the answer in advance.
3. Quantify how priority, exam type, shift, scanner, patient setting and contrast status affect the process.
4. Turn all of that into recommendations somebody could actually act on.

The condition that changed everything

One rule was agreed at kickoff and it shaped every decision that followed. No bottleneck would be assumed before analysis. Every stage would be measured independently and the data would decide.

That sounds like a small methodological nicety. It is not. The intuitive answer, that a busy scanner is the constraint, is where almost everyone starts, and in this dataset it is wrong. Committing to measure everything is the only reason the real answer surfaced.

2. Data and Method

The dataset

20,075 CT examination records covering 1 January to 31 December 2025. Seven workflow timestamps per exam, plus the exam and operational attributes that might explain the differences between them.

Field group	Fields
Workflow timestamps	Order, Scheduled, Arrival, Scan Start, Scan End, Report Start, Report Finalized
Exam attributes	Exam type, priority, contrast status
Operational context	Patient setting, scanner, shift, radiologist, technologist, status
Identifiers	Exam ID, patient ID

Six scripts, each with one job

Script	What it does
01 Ingestion	A staging table where nothing can be rejected, then a typed load that loses nothing
02 Profiling	Counts, duplicates, blanks, category audit, chronology checks, all before anything changes
03 Cleaning	De-duplication and category standardization
04 Metrics	Six stage durations, total turnaround, and a data quality flag
05 KPIs	Headline numbers and the agreed business questions
06 Advanced	Variance decomposition, confounding checks, effect sizes, tiered SLA

Cutting the workflow into six pieces

The CT process splits into six intervals that follow one another in a fixed order and add up to the whole journey.

Stage	From, to	What it really is
1 Order to Schedule	Order, Scheduled	An administrative queue
2 Schedule to Arrival	Scheduled, Arrival	Patient-side lead time
3 Arrival to Scan	Arrival, Scan Start	Waiting in the department
4 Scan Duration	Scan Start, Scan End	The scan itself
5 Scan to Report Start	Scan End, Report Start	Handoff to the radiologist
6 Reporting Duration	Report Start, Report Finalized	Reading and signing off

Because the six stages are mutually exclusive and add up to total turnaround, the variance of turnaround can be split across them. That property is what makes the central finding in Section 4.1 possible, and it is why every stage had to be measured rather than estimated.

Why the dashboard reads a view and not a table

Power BI connects to one curated SQL view, `vw_ct_kpi_ready`, and nothing else. De-duplication, standardization, validation, population rules and duration logic are all settled in SQL before a single number reaches the dashboard.

The dashboard shows; SQL decides. That means the definition of a completed, valid, reportable exam lives in one version-controlled place instead of being reinvented in DAX every time somebody builds a new visual.

3. What Was Wrong With the Data

Before any analysis, the dataset had to be trusted. At first it could not be.

3.1 The load that lied

Severity: critical. Nothing reported an error.

The first import brought in 20,015 rows. The source file has 20,075.

Sixty records were gone, and the import wizard said everything went fine.

The cause was one line in the table definition. The report-finalized column had been declared as a datetime type. Sixty exams legitimately had that field blank, because the report was never signed off. Rather than storing the blank as null, the import threw away the entire row.

Here is why that matters more than it looks. Run this on the truncated table:

```
SELECT COUNT(*) FROM ct_imaging_raw WHERE Report_Finalized_Time IS NULL;
```

It returns zero.

Zero unfinalized reports. A department with a flawless completion rate. Every single record capable of proving otherwise had already been deleted by the tool that was supposed to load them, silently, with no warning and no failed row count on screen.

It surfaced for one reason only. The loaded row count was compared against the source file. Nothing in the software output gave it away.

The fix was to rebuild the load in two layers. First a staging table where every column is plain text, so nothing can be rejected on type. Then a typed conversion under explicit control, mapping blanks to null instead of discarding the row. All 20,075 records now arrive, and the missing report timestamps survive where they can be counted and reasoned about.

Row-count reconciliation now runs on every load, before anything downstream. It is the cheapest check in the pipeline and it caught the most expensive problem in it.

3.2 Everything else that was wrong with it

Profiling came before cleaning, always. Every transformation below is justified by a measurement taken first.

What was wrong	How many	What was done
The same Exam_ID appearing twice	75	Window function, earliest order kept
Priority spelled six ways for three categories	219	Standardized to Routine, Urgent, STAT
Contrast spelled four ways for two	181	Trimmed and standardized to Yes, No
Blank Contrast_Status	119	Kept, shown as Unknown
Blank Scanner_ID	80	Kept, shown as Unknown
Blank Patient_Setting	80	Kept, shown as Unknown
Blank Report_Finalized_Time	60	Kept as null, left out of turnaround
Blank Priority	59	Kept, shown as Unknown
Scheduled before it was ordered	35	Flagged Invalid

What was wrong	How many	What was done
Scan ended before it started	80	Flagged Invalid
Report started before the scan finished	60	Flagged Invalid

All counts above are measured across the full 20,000 de-duplicated records, before any filtering on exam status. The three chronology checks overlap, so they cover 174 distinct records rather than 175. The category problems are the sneaky kind. Priority arrived as Routine, routine, Urgent, URGENT, STAT and Stat. Contrast status included the word No with a trailing space, which looks identical to No in any result grid you will ever open. Left alone, every grouped query would have split each real category across several rows and quietly understated all of them.

The duplicates turned out to be reassuring rather than alarming. All 75 pairs are identical across all 18 fields and share the same order timestamp. They are load artifacts, not two competing versions of the truth, which means the tie-break rule used to remove them is arbitrary and provably safe, because either copy gives the same answer.

3.3 Two decisions worth defending

Records with missing categories were kept, not deleted. The tempting move is to drop them, because it makes the data tidier. It is also quietly dishonest, because these are real completed examinations. Removing them pulls genuine exams out of every volume count and shifts every average in a direction nobody can audit later. They appear instead as an explicit Unknown category, so the gap in the data is something a reader can see rather than something that silently moved the numbers.

Impossible timestamps were flagged, not fixed. There is no honest way to invent a timestamp. Any repair would be a made-up value wearing the costume of a measurement, and six months later nobody could tell it apart from a real one. Flagging keeps the exclusion auditable: the count is queryable at any time and the decision can be revisited without reloading anything.

3.4 Where all 20,075 rows ended up

Step	Records	Lost	Why
Loaded from source	20,075		
After removing duplicates	20,000	75	Repeated Exam_ID
Completed exams only	19,206	794	400 canceled, 394 no-show
Passing the chronology checks	19,037	169	Flagged Invalid
With a finalized report	18,981	56	Report never signed off

Each row shows what that step removed from the records still standing at that point. That is why the chronology step shows 169 rather than the 174 broken records counted in 3.2: five of those 174 are no-shows or cancellations, already removed one step earlier. Every excluded record is accounted for. The 794 canceled and no-show exams cannot have a turnaround time, so they sit outside the KPIs, but they consumed booked scanner capacity and produced nothing clinical. That is a real cost turnaround metrics are structurally incapable of seeing, which is why it gets its own reporting line rather than being quietly filtered away.

4. What the Data Said

4.1 Two stages cause 86% of the variation

This is where the project earns its keep.

Ranking stages by average length tells you where the time sits. It does not tell you why one exam takes 100 minutes and the next takes 250. A stage can be long and completely predictable, which makes it a fixed cost rather than a source of chaos, and fixed costs are not where you spend improvement money.

Because the six stages add up to total turnaround, the variance of turnaround can be split across them. That is the calculation that answers the real question.

Stage	Average (min)	Std dev	Share of average TAT	Share of the variance
Order to Schedule	56.8	38.6	32.0%	50.4%
Reporting	43.6	32.3	24.6%	35.4%
Schedule to Arrival	33.7	17.7	19.0%	10.1%
Scan duration	22.7	8.3	12.8%	2.5%
Scan to Report start	10.1	5.5	5.7%	1.1%
Arrival to Scan	8.0	3.7	4.5%	0.4%

The shares sum to exactly 100%, which is the built-in proof the decomposition is correct.

Waiting for an order to be scheduled and waiting for a report to be written explain 86% of the variation between a fast exam and a slow one.

Now look at the bottom three rows. Patient wait, the scan itself, and the handoff to reporting together take 24% of the average and produce 4% of the variance. They are quick, and they are consistent.

The scanner is not the problem.

Turnaround is decided before the patient ever walks in and after they walk out. The machine in the middle is doing its job. This is the finding the department would never have reached by looking at scanner utilization, which is where almost everyone looks first.

4.2 Priority works, exactly where you would hope

Priority	Exams	Avg TAT	Order to Schedule	Reporting	Scan duration
STAT	2,850	153.2	39.5	36.6	22.8
Urgent	5,692	172.5	53.2	42.5	22.5
Routine	10,385	186.7	63.5	46.1	22.7

STAT exams finish 33.5 minutes ahead of routine ones. Almost all of that gap comes from the two variable stages: 24.0 minutes of faster scheduling and 9.5 minutes of faster reporting.

Scan duration barely moves, 22.5 to 22.8 minutes across all three priorities. That is the right answer and a useful sanity check on the whole measurement approach. Marking an exam STAT moves it up the queue. It does not make the scanner spin faster.

4.3 A finding I had to throw away

Contrast-enhanced exams average 182.2 minutes against 173.4 without contrast. An 8.8 minute penalty. It would be easy, and wrong, to write a recommendation about contrast workflow on the back of that.

Contrast is used in roughly 72% of CT Angiography, Abdomen/Pelvis and Chest exams, and only about 12% of Head, Spine and Extremity exams. The contrast group is stacked with the exam types that were always going to be slow. The comparison is partly measuring exam mix wearing a contrast label.

Splitting it out by exam type:

Exam type	No contrast	With contrast	Difference	% using contrast
Extremity CT	170.9	181.3	+10.4	12.6%
Chest CT	171.0	178.5	+7.6	71.5%
Abdomen/Pelvis CT	172.5	178.6	+6.1	71.9%
Spine CT	172.3	178.0	+5.8	11.6%
Head CT	172.3	176.2	+3.9	11.7%
CT Angiography	197.5	197.9	+0.3	72.5%

Look at the bottom row. Within CT Angiography, the exam type that uses the most contrast and takes the longest, contrast makes no measurable difference at all.

Conclusion

Contrast is not an independent driver of turnaround at anything like the size the raw comparison suggests. Acting on that 8.8 minute figure would have meant chasing a confounder and spending clinical goodwill on a change that would not have moved the number.

4.4 The difference that was not worth reporting

Scanner CT-03 averages 181.9 minutes; CT-02 averages 173.8. An eight minute spread, against a standard deviation of 55 minutes. With roughly 4,500 exams per scanner that difference is statistically detectable, and the effect size runs from minus 0.06 to plus 0.08, which is nothing.

Factor	Range	Spread	Verdict
Scanner	CT-02 173.8 to CT-03 181.9	8.1 min	Noise. Not worth acting on.
Shift	Day 170.2 to Night 182.3	12.1 min	Real, consistent, worth watching.
Exam type	Extremity 172.1 to Angiography 197.8	25.7 min	Longer acquisition and reading. Clinical complexity, not a process fault.

Reporting the non-finding is the useful thing to do. Telling a department to investigate CT-03 on that evidence would send people looking for a problem that is not there. Not every difference that reaches statistical significance deserves somebody time.

4.5 The SLA was measuring the wrong thing

Against a flat 120-minute target, compliance is 12.2%.

Read on its own, that number describes a department in freefall. It is not a useful measurement. It holds routine outpatient imaging to an emergency standard and then reports the predictable result.

Give each priority a target that matches its clinical urgency and the picture changes completely.

Priority	Target	Exams	Within target	Compliance
STAT	60 min	2,850	13	0.5%
Urgent	120 min	5,692	781	13.7%
Routine	240 min	10,385	8,897	85.7%
Blended		18,927	9,691	51.2%

Conclusion

Routine imaging is broadly meeting a sensible expectation. Urgent and STAT are not. The department does not have a general slowness problem; it has a specific one, where clinical urgency is not translating into compressed turnaround exactly where it matters most. Which points straight back to Section 4.1, because the STAT advantage that does exist is mostly faster scheduling.

Both thresholds are illustrative assumptions for a synthetic dataset, not published clinical standards. A real engagement would set them with radiology and the emergency department in the room.

4.6 The same answer, arrived at a different way

Comparing exams that met the 120-minute target against those that missed it:

SLA status	Total TAT	Order Schedule to	Schedule to Arrival	Arrival to Scan	Scan	Scan Report to	Reporting
Within SLA	104.4	22.3	24.0	7.5	19.4	8.5	20.2
Breached	187.5	61.6	35.1	8.1	23.1	10.3	46.8

SLA status	Total TAT	Order Schedule to	Schedule to Arrival	Arrival to Scan	Scan	Scan Report to	Reporting
Difference	+83.1	+39.3	+11.1	+0.6	+3.7	+1.8	+26.6

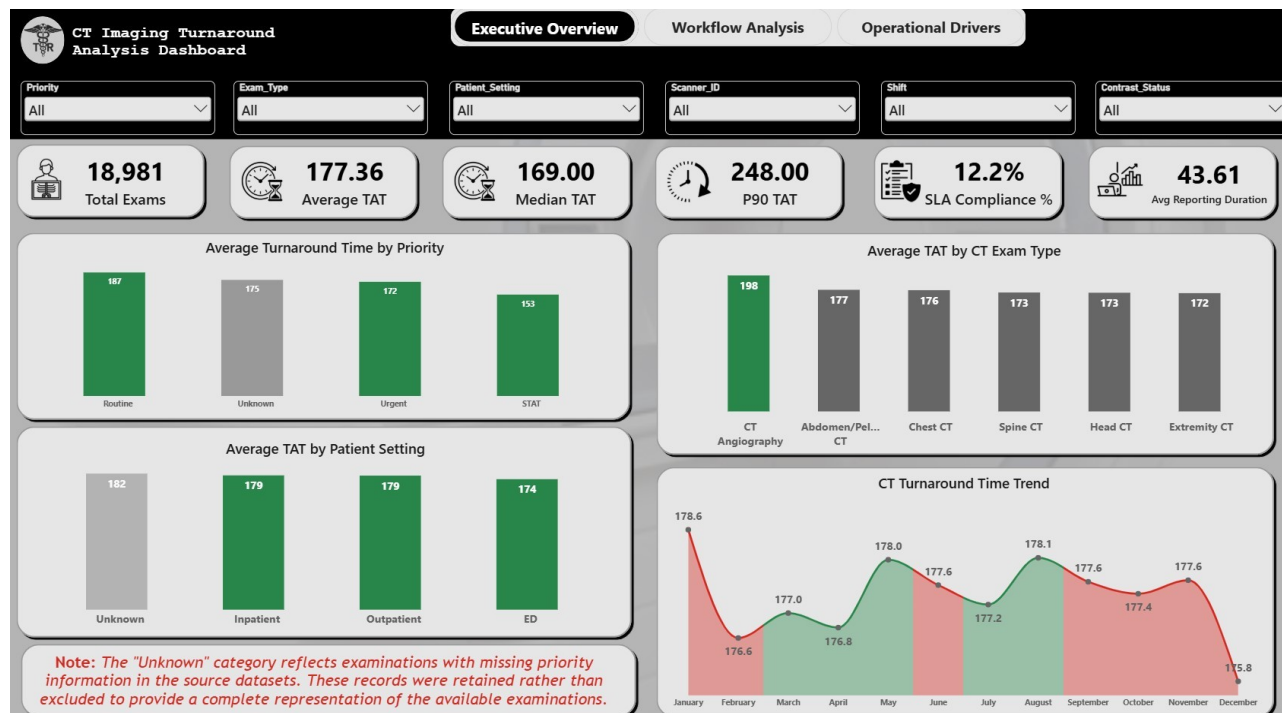
Of the 83-minute gap between a compliant exam and a breached one, 79% comes from two stages: waiting to be scheduled, and waiting to be read.

Arrival to scan differs by 0.6 minutes. The imaging suite performs essentially identically whether an exam hits its target or misses it by an hour.

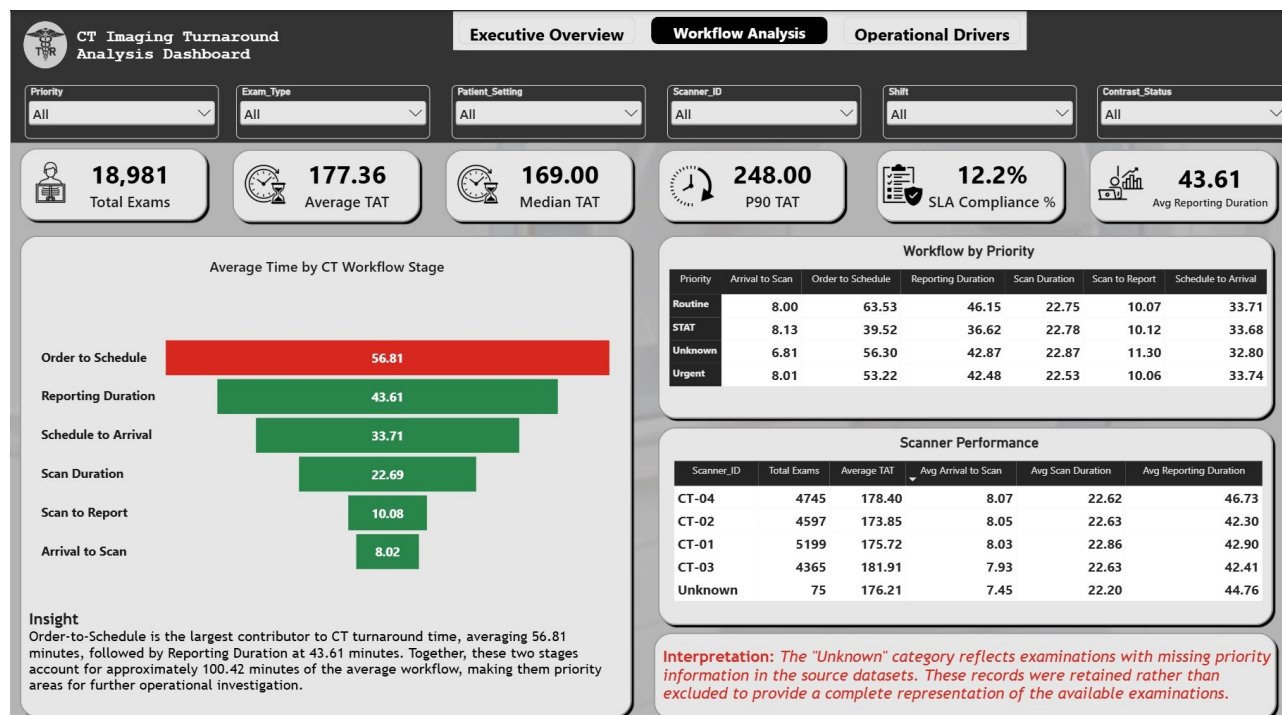
This is a completely separate calculation from the variance decomposition, and it lands in the same place. When two independent routes agree, the finding is solid.

5. The Dashboard

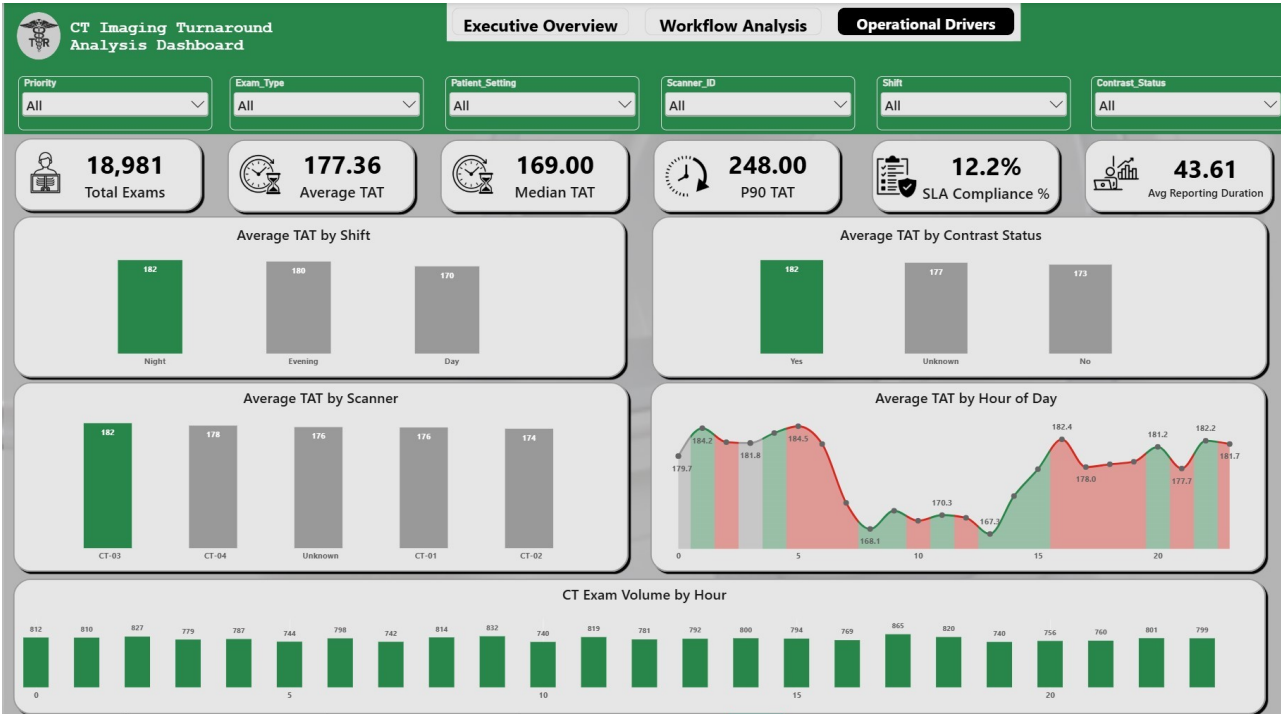
Three pages in Power BI, cross-filtered on priority, exam type, patient setting, scanner, shift and contrast status. Every page reads the same curated SQL view, so no number on screen was calculated twice in two places.



Page 1, Executive Overview. Headline KPIs, turnaround by priority, exam type and patient setting, and the monthly trend. The note explains why Unknown appears as a category rather than being filtered away.



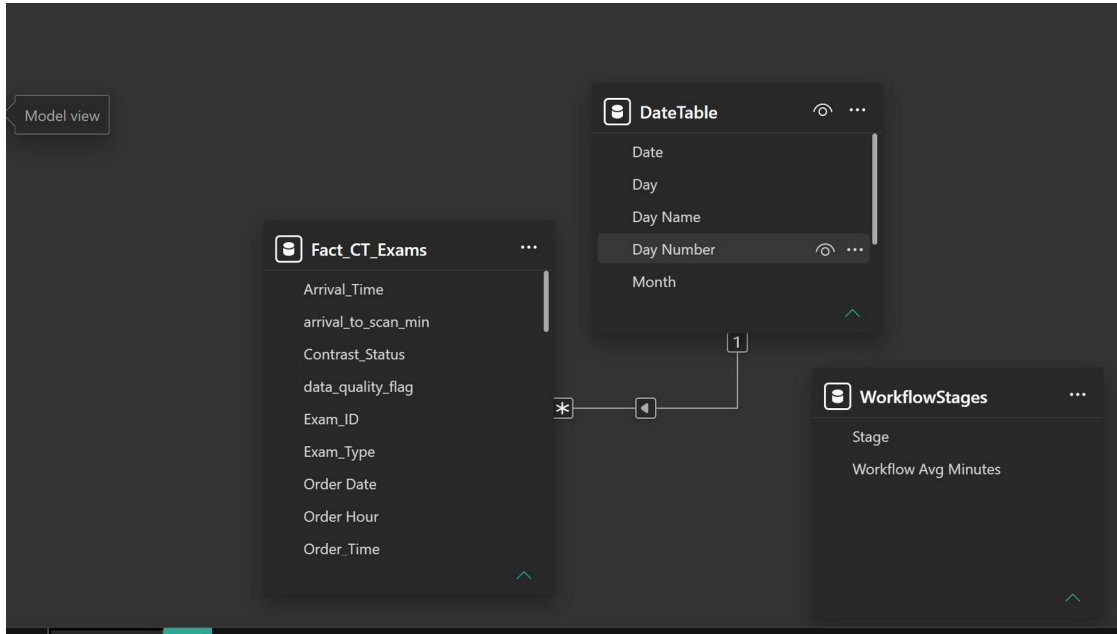
Page 2, Workflow Analysis. Average time by stage, the workflow broken down by priority, and scanner performance including reporting duration.



Page 3, Operational Drivers. Turnaround by shift, contrast status, scanner and hour of day, with exam volume by hour underneath.

5.1 The data model behind it

A star schema. Fact_CT_Exams holds one row per examination with its stage durations already computed in SQL. DataTable is a proper date dimension built in DAX, joined one-to-many on order date, which is what makes the time intelligence work. WorkflowStages holds the six stage names so the stage comparison can be rendered as a single ranked visual rather than six unrelated cards.



Power BI model view. Durations are calculated upstream in SQL, so the model stays thin and the measures stay simple.

6. What I Would Tell the Department

1. Start with order-to-schedule.

It is the longest stage at 56.8 minutes, the single biggest source of variation at 50%, and it has the widest tail of anything in the workflow, with a 90th percentile of 105 minutes. It is also administrative rather than clinical, which makes it the cheapest thing on this list to change. Auto-scheduling protocol-eligible routine exams, and routing STAT orders straight into a held slot, go after the biggest lever first.

2. Then go after reporting turnaround.

It averages 43.6 minutes with the widest spread of anything after the scan, a standard deviation of 32.3 and a 90th percentile of 83 minutes, and it contributes 35% of total variance. Worklist prioritization by clinical urgency, plus a look at radiologist coverage against evening and night volume, both act here.

3. Do not spend money on scanner capacity.

Arrival to scan at 8.0 minutes, scan duration at 22.7 and the handoff to reporting at 10.1 are fast and tightly clustered, and together they account for 4% of the variance. A new scanner would not move the headline number. This is the recommendation that saves the most money, and it is the one that would never have been made without measuring every stage.

4. Replace the flat SLA with tiered targets.

The single 120-minute threshold reports 12.2% compliance and hides the fact that routine imaging is broadly fine while STAT is not. Tiered targets put attention where the actual gap is, instead of spreading alarm evenly across a department that is only failing in one place.

5. Report the demand-loss rate.

794 canceled and no-show exams, 3.97% of all orders, consumed booked capacity and produced nothing clinical. Turnaround metrics cannot see this by construction, so it needs its own line.

6. Reconcile row counts on every load.

The failure that quietly deleted 60 records raised no error and would have reported a perfect completion rate. One count, run every time, is the whole fix.

7. What This Analysis Cannot Tell You

- The data is synthetic. The relationships in it were generated, not observed. The method transfers to real data. The specific numbers do not.
- Exam volume is nearly flat across all 24 hours, which no real radiology department has ever looked like. Turnaround by hour is worth reading. Volume by hour is an artifact of how the data was made.
- The SLA thresholds are assumptions, chosen to be reasonable, not standards.
- 174 records fail at least one chronology check across the full dataset. 169 of those are completed exams and were excluded from the KPIs on that basis; the other five are no-shows and cancellations, already removed by the status filter. Both figures are correct and count different populations. All are broken by construction, so nothing was imputed.
- Minute arithmetic truncates toward zero. Because each of the six stages truncates separately while total turnaround truncates once, the stage averages sum to 174.9 against a measured average of 177.4. Expected rather than an error, and documented so the 2.4 minute gap has an answer ready.
- This is descriptive analysis. Nothing here is causal, and Section 4.3 is a live demonstration of why differences in this data should not be read that way.
- Radiologist reporting times are not adjusted for case mix. Somebody reading more CT Angiography will look slower for reasons of complexity, not performance, so those figures are not a ranking of anybody.

8. Reproducing the Work

Tool	What it did
MySQL 8.0	Ingestion, profiling, cleaning, metrics, analysis
Power BI	Data model, DAX measures, three-page dashboard
Excel	Independent validation of every KPI, and field documentation

Everything is in the project repository: the six SQL scripts, the raw dataset, the data dictionary, the Power BI file, the Excel validation workbook, the data quality log, and step-by-step build notes. The analysis reproduces end to end from the raw CSV.

You will need MySQL 8.0 or later, because the analysis uses window functions. Check that 20,075 rows loaded before running the profiling script. That one check is what catches the ingestion failure described in Section 3.1.